

Yukawa generative models

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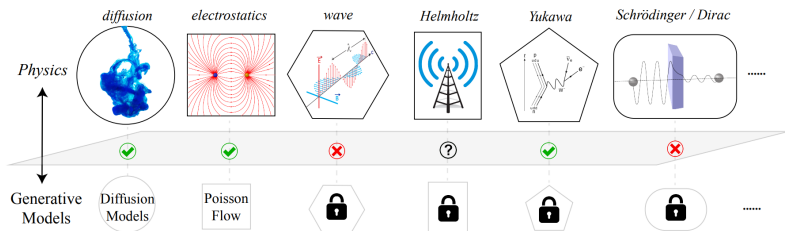
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GenPhys approach



Interpretable physics-inspired generative models

Thermodynamics



Diffusion models

Electrostatics



PFGM

EFM

Diffusion-based models

Diffusion Models (DM) are based on thermodynamics and described by the forward and backward dynamics as:

$$dx_t = f(x_t, t) dt + g(t) dw_t \quad (1)$$

$$dx_t = (f(x_t, t) - g^2(t) s_\theta(x_t, t)) dt + g(t) d\bar{w}_t \quad (2)$$

where $s_\theta(x_t, t)$ is an approximation of $\nabla_x \log p_t(x_t)$.

The **forward** process (1) describes a diffusion stochastic process that corrupts data, injecting Gaussian pure noise.

The **backward** process (2) describes a diffusion stochastic process that recovers data from Gaussian noise.



Maxwell's electrostatics

The point charge q at a point $\mathbf{x}' \in \mathbb{R}^{D+1}$ creates field:

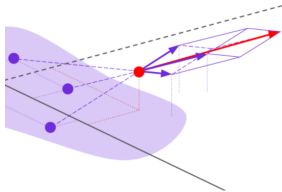
$$\mathbf{E}(\mathbf{x}) = \frac{q}{S_D} \frac{\mathbf{x} - \mathbf{x}'}{\|\mathbf{x} - \mathbf{x}'\|^{D+1}}. \quad (3)$$

If there are some charges $q_1, \dots, q_N$ at points $\mathbf{x}_1, \dots, \mathbf{x}_N$, they create fields $\mathbf{E}_1(\mathbf{x}), \dots, \mathbf{E}_N(\mathbf{x})$ at $\mathbf{x} \in \mathbb{R}^{D+1}$. The total field at $\mathbf{x}$ is given by the superposition principle, where S_D is an area of D -dimensional sphere:

$$\mathbf{E}(\mathbf{x}) = \sum_{n=1}^N \mathbf{E}_n(\mathbf{x}) = \sum_{n=1}^N \frac{q_n}{S_D} \frac{\mathbf{x} - \mathbf{x}_n}{\|\mathbf{x} - \mathbf{x}_n\|^{D+1}}. \quad (4)$$

The electric field of continuously distributed charge $q(\mathbf{x})$:

$$\mathbf{E}(\mathbf{x}) = \int \frac{1}{S_D} \frac{\mathbf{x} - \mathbf{x}'}{\|\mathbf{x} - \mathbf{x}'\|^{D+1}} q(\mathbf{x}') d\mathbf{x}'. \quad (5)$$



Maxwell's electrostatics

The field flux of $\mathbf{E}(\mathbf{x})$ through a surface Σ is defined as:

$$\Phi = \iint_{\Sigma} d\Phi = \iint_{\Sigma} \mathbf{E} \cdot d\mathbf{S}. \quad (6)$$

where $d\mathbf{S}$ is an area element of the finite surface Σ .

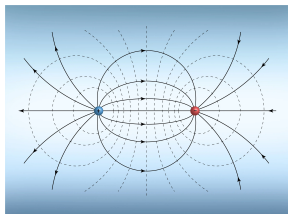
Intuition: The flux of electric field Φ indicates the density of field lines passing through the finite surface Σ .

An electric field strength line

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{E}(\mathbf{x}(t))$$

Potential

$$\mathbf{E}(\mathbf{x}) = -\nabla\varphi(\mathbf{x})$$



Electrostatic Field Matching (EFM)

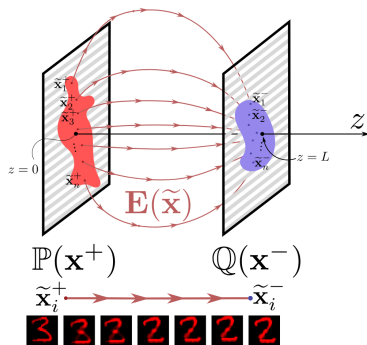
Distributions are considered as charge densities and assign:

- ▶ $P(\mathbf{x}^+)$ as a distribution of positive charges, $\mathbf{x}^+ \in \mathbb{R}^D$.
- ▶ $Q(\mathbf{x}^-)$ as a distribution of negative charges, $\mathbf{x}^- \in \mathbb{R}^D$.

$\mathbf{x}^+$ and $\mathbf{x}^-$ are augmented with $z = 0$ and $z = L$:

$$q^+(\tilde{\mathbf{x}}) = q^+(\mathbf{x}, z) = P(\mathbf{x})\delta(z),$$

$$q^-(\tilde{\mathbf{x}}) = q^-(\mathbf{x}, z) = -Q(\mathbf{x})\delta(z - L). \quad (9)$$



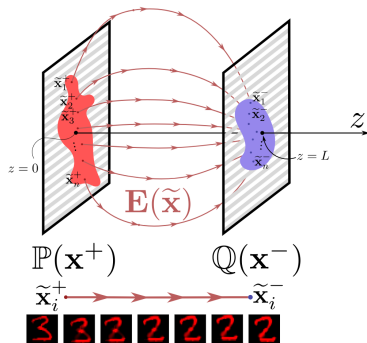
Electrostatic Field Matching (EFM): Field

Since each plate creates its own field at a point $\tilde{\mathbf{x}} \in \mathbb{R}^{D+1}$, the superposition principle is used to define the total field:

$$\mathbf{E}(\tilde{\mathbf{x}}) = \mathbf{E}^+(\tilde{\mathbf{x}}) + \mathbf{E}^-(\tilde{\mathbf{x}}). \quad (10)$$

In accordance with Coulomb's law, the exact expression of each summand is:

$$\mathbf{E}^\pm(\tilde{\mathbf{x}}) = \int \frac{1}{S_D} \frac{\tilde{\mathbf{x}} - \tilde{\mathbf{x}}'}{\|\tilde{\mathbf{x}} - \tilde{\mathbf{x}}'\|^{D+1}} q^\pm(\tilde{\mathbf{x}}') \quad (11)$$



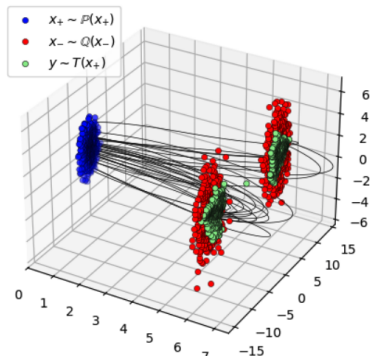
Electrostatic Field Matching: Limitations

Backward-oriented field lines:

Each plate produces two sets of electric field lines. The use of only forward-oriented lines is not sufficient to fully cover the target.

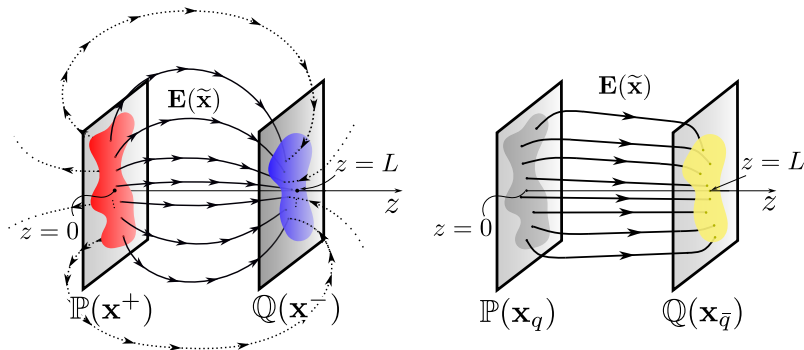
Line termination problem: Some forward-oriented lines enter the region $z > L$ and require further integration.

Training volume selection: It is necessary to know not only the field between the plates, but also beyond.



Existing approaches to handle these limitations

Interaction field matching (IFM)



Interaction Field Matching: Overcoming Limitations of Electrostatic Models (Manukhov et al.)

Interaction Field Matching

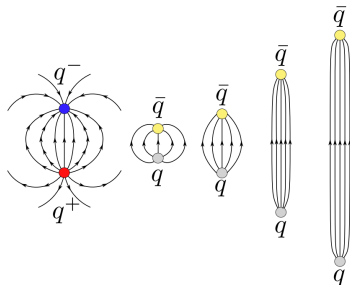
To tackle the challenge, IFM proposes to consider **strong interaction** between quark $\tilde{\mathbf{x}}_q$ and anti-quark $\tilde{\mathbf{x}}_{\bar{q}}$ in $\mathbb{R}^{D+1}$.

The field of strong interaction has the next properties:

Low curvature of field lines: The separation increases, the strong field lines become considerably straighter.

Flux conservation: Interaction field $\mathbf{E}$ maintains the preserving of flux along the stream tube: $\mathbf{E}(\tilde{\mathbf{x}}) \cdot d\mathbf{S} = \text{const.}$

Overcoming EFM limitations: There are no backward-oriented lines or lines beyond $z > L$. Training volume.



Yukawa interaction

Poisson and Yukawa comparison

	Poisson	Screened Poisson (Yukawa)
Equation	$\nabla^2 \varphi = f$	$\nabla^2 \varphi - m\varphi = f$
Solution for 3D ($f = 0$)	$\varphi \sim \frac{1}{r}$	$\varphi \sim \frac{e^{-mr}}{r}$

Yukawa potential has «finite interaction radius» $R \approx \frac{1}{m}$

Green function technique

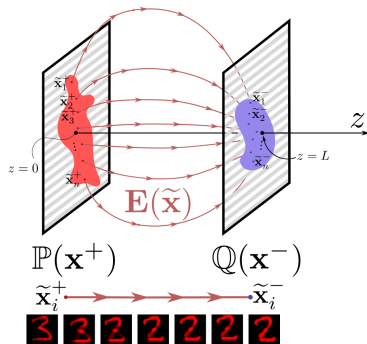
$$\varphi(x, t) = \int G(x, t; x', t') f(x', t') dx' dt'.$$

$$G_{\text{scr}}(r, t) = \frac{1}{(2\pi)^{\frac{N+1}{2}}} \left(\frac{m}{\sqrt{t^2 + r^2}} \right)^{\frac{N-1}{2}} K_{\frac{N-1}{2}}(m\sqrt{t^2 + r^2}),$$

with K_ν the modified Bessel function of the second kind, $r = \|x - x'\|$, $t' = 0$.

Yukawa generative model

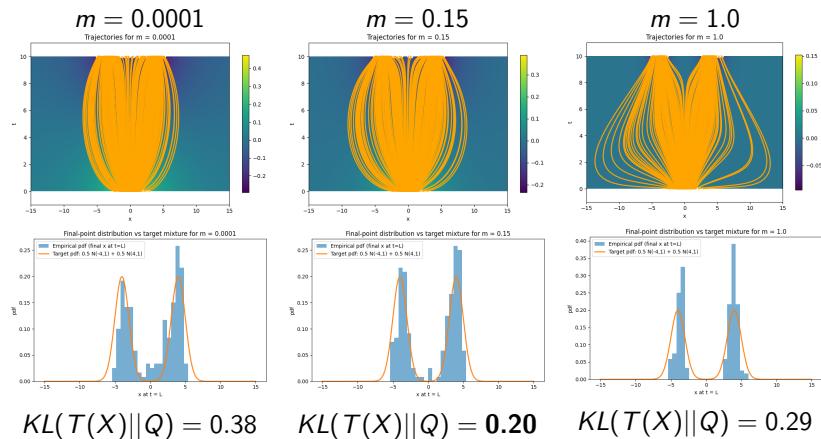
1. Using the Green function, calculate the potentials in the training volume created by positive charges on the left plate of the capacitor and negative charges on the right plate of the capacitor.
2. Using $\mathbf{E}(\mathbf{x}) = -\nabla\varphi(\mathbf{x})$ learn electric field strength lines
3. Verify that massless particles x from the left distribution will indeed enter the right distribution $T(X) \approx Q$ if they move along the force lines



1D toy experiments. Trajectories

Hypothesis: the best interaction radius $R = \frac{1}{m} \approx L$.

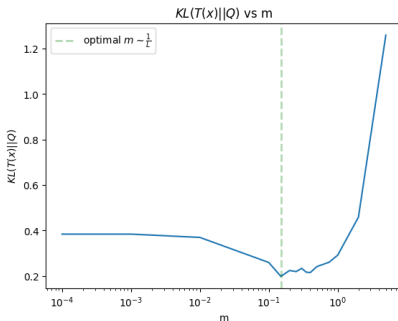
In our experiment $L = 10$, therefore optimal m should be approximately $m \approx \frac{1}{L} = 0.1$



1D toy experiment. Conclusion

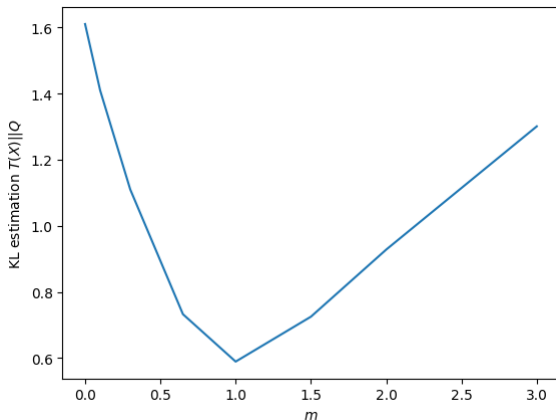
Results:

1. Small m values correspond to the EFM method with its problems
2. Large values of m strongly attract particles to their modes, without covering the entire support of the Q distribution
3. The optimal value of m that ensures minimal KL-divergence $KL(T(X)||Q)$ is equal to $m \approx \frac{1}{L}$



2D toy experiment: Gaussian $\rightarrow$ Mixture of Gaussians

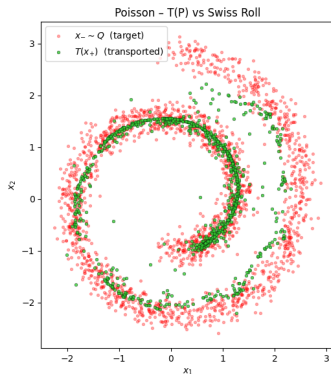
The same result for 2D toy experiment: there is optimal m which gives the minimal KL-divergence



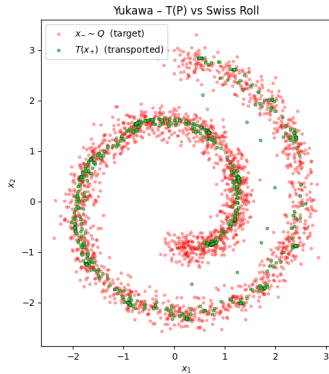
2D toy experiment: Gaussian $\rightarrow$ SwissRoll

The generative model based on Yukawa's potential visually reproduces the target SwissRoll better than the EFM

Poisson potential (EFM)



Yukawa potential



Conclusion

Takeaways

Extending the EFM method to the case of the Yukawa field improves the solution of the domain translation problem

Future plans

1. Conduct experiments on Colored MNIST, CIFAR-10, etc.
2. Theoretically explain the observed improvement in domain translation task solution
3. To theoretically estimate the optimal value of m , in particular, how it depends on the dimensionality of the space

Physical processes and physical PDEs

$$(L\varphi)(x, t) = F(\varphi, \varphi_t, \varphi_{tt}, \nabla\varphi, \nabla^2\varphi, \dots) = f(x, t),$$

Green's function G is defined by

$$(LG)(x, t; x', t') = \delta(x - x') \delta(t - t'),$$

Then the solution is the convolution

$$\varphi(x, t) = \int G(x, t; x', t') f(x', t') dx' dt'.$$

1) Diffusion equation :

$$\varphi_t - \nabla^2\varphi = 0, \quad G_{\text{diff}}(r, t) = \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{r^2}{4t}\right).$$

2) Poisson equation :

$$\varphi_{tt} + \nabla^2\varphi = 0, \quad G_{\text{Pois}}(r, t) = \frac{1}{(t^2 + r^2)^{\frac{N-1}{2}}},$$

3) Screened Poisson (Yukawa) equation :

$$\varphi_{tt} + (\nabla^2 - m^2)\varphi = 0$$

$$G_{\text{scr}}(r, t) = \frac{1}{(2\pi)^{(N+1)/2}} \left(\frac{m}{\sqrt{t^2 + r^2}} \right)^{(N-1)/2} K_{(N-1)/2}(m\sqrt{t^2 + r^2}),$$

with K_ν the modified Bessel function of the second kind